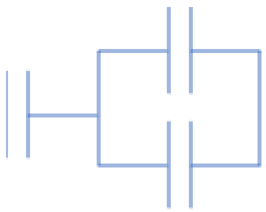
SECTION A

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		Capacitance equation used i.e. $C = \frac{\varepsilon_0 A}{d}$ (1) Area = d^2 and cm conversion (7.569×10^{-3} m^2) (1) Correct separation (2.91×10^{-4} m or 0.291 mm etc.) (1)	1	1 1		3	2	
	(b)		Using $Q = CV$ e.g. 87 [V] **or** substituting (1) Using the energy equation i.e. $\frac{Q^2}{2C}$ (1st mark implied) (1) Final answer = 2.035×10^{-8} [C] or 20.35 n[C] **hence** correct (1) e.g. 0.87 μJ $\approx$ 0.9 μJ Accept 0.87 μ[J], 222 p[F]			3	3	3	
	(c)		Same pd or full pd [to each C] (1) Same charge [on each C] **OR** $C = \frac{Q_{\text{total}}}{V} = \frac{Q+Q}{V}$ **OR** $C = \frac{Q_1 + Q_2}{V}$ (1) Hence double charge **OR** $C = \frac{Q}{V} + \frac{Q}{V}$ **OR** $C = \frac{Q_1}{V} + \frac{Q_2}{V}$ i.e. seeing that capacitances can be added (1)		3		3		
	(d)		2×230 pF in parallel and then 230 pF in series 		1		1		
			Question 1 total	**1**	**6**	**3**	**10**	**5**	**0**